WORKED SOLUTION

STEP II 2014, Question 7

Pure • Curve sketching

> **STEP II 2014, Question 7**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/14-s2-q7

The key idea

A sum of modulus functions is piecewise linear: it changes gradient only at the points where one of the moduli is zero. So the graph of $|x-a|+|x-b|$ is a "bath" with a flat floor between a and b and walls of gradient ± 2. Counting solutions of the later equations then reduces to comparing gradients: a graph of gradient 2 always overtakes one of gradient 1, and two graphs with the same gradient either coincide or never meet.

Solution

Throughout, write $f(x) = |x-a|+|x-b|$ with $a < b$.

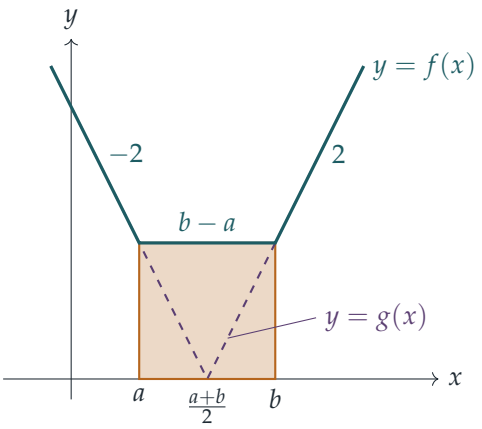
Part (i)

Removing the moduli region by region:

$$f(x) = \begin{cases} a+b-2x, & x \le a, \\ b-a, & a \le x \le b, \\ 2x-a-b, & x \ge b. \end{cases}$$

So the gradient is -2 for $x < a$, 0 for $a < x < b$ and 2 for $x > b$. The graph is continuous, with a flat floor at height $b-a$ above $[a,b]$.

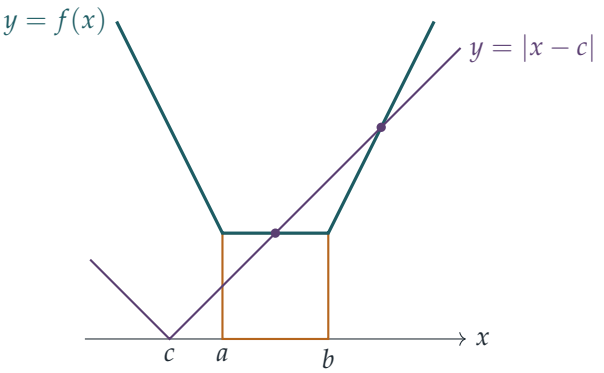
The graph of $g(x) = |2x-a-b| = 2|x-\frac{a+b}{2}|$ is a V with vertex at $(\frac{a+b}{2}, 0)$ and gradients ± 2. For $x \ge b$ we have $2x-a-b \ge b-a > 0$, so $g(x) = 2x-a-b = f(x)$, and similarly $g(x) = a+b-2x = f(x)$ for $x \le a$. Hence the two graphs coincide outside (a,b). Inside, g dips down to 0 while f stays at $b-a$. (In fact $f(x) = \max\{b-a,\, g(x)\}$.)



The quadrilateral has vertices $(a,0)$, $(b,0)$, $(b,b-a)$ and $(a,b-a)$, using $f(a)=f(b)=b-a$. Its base has length $b-a$, its vertical sides have length $b-a$, and all its angles are right angles. So it is a **square** of side $b-a$. Notice too that the V of g runs from the midpoint of the square's base up through its two top corners, since $g(a)=g(b)=b-a$.

Part (ii)

Now compare $y=f(x)$ with $y=|x-c|$, a V with vertex $(c,0)$ on the axis and arms of gradient ±1. The key fact is that f is always at least $b-a>0$, and its walls (gradient ±2) are steeper than the arms of the V. The sketch shows the case $c<a$, where the V cuts the bath twice.



Case $a<c<b$: no solutions. On $[a,b]$, the distance $|x-c|$ is at most $\max\{c-a,\,b-c\}<b-a=f(x)$, so the V stays strictly below the floor of the bath. For $x\geq b$,

$$f(x)-|x-c|=(2x-a-b)-(x-c)=x-b+(c-a)\geq c-a>0,$$

and the same argument (or the symmetry below) deals with $x\leq a$. So the V lies entirely inside the bath, and there are no solutions.

Case $c=a$ or $c=b$: one solution. Take $c=a$. On $[a,b]$ we have $|x-a|=x-a\leq b-a$, with equality only at $x=b$. For $x>b$, $f(x)-(x-a)=x-b>0$; for $x<a$,

$f(x) - (a - x) = b - x > 0$. So the only solution is $x = b$: the right arm of the V passes exactly through the top corner $(b, b - a)$ of the square. Similarly, when $c = b$ the only solution is $x = a$.

Case $c < a$: two solutions. The left arm $y = c - x$ lives on $x \leq c < a$, where f has gradient -2. At $x = c$ we have $f(c) > 0 = c - c$, and as we move left f rises faster than the arm, so $f(x) > c - x$ for all $x \leq c$. The left arm gives no solutions.

For the right arm $y = x - c$ ($x \geq c$), let $k(x) = f(x) - (x - c)$. Subtracting 1 from the gradients of f, k has gradient -3 on (c, a), -1 on (a, b) and $+1$ for $x > b$. So k is strictly decreasing on $[c, b]$ and strictly increasing on $[b, \infty)$. Also

$$k(c) = f(c) > 0, \qquad k(b) = (b - a) - (b - c) = c - a < 0, \qquad k(x) \to \infty \text{ as } x \to \infty.$$

Hence k has exactly one zero in (c, b) and exactly one in (b, ∞). Explicitly, these are

$$x = a + b - c \quad \text{and} \quad x = \begin{cases} b - a + c, & 2a - b \leq c < a, \\ \frac{1}{3}(a + b + c), & c < 2a - b, \end{cases}$$

the second one meeting the floor of the bath or its left wall respectively.

Case $c > b$: two solutions. The reflection $x \mapsto a + b - x$ in the line $x = \frac{a+b}{2}$ leaves f unchanged (it swaps $|x - a|$ and $|x - b|$) and sends $|x - c|$ to $|x - c'|$ with $c' = a + b - c < a$. So this case is the mirror image of the previous one.

In summary, there are 0 solutions if $a < c < b$, 1 solution if $c = a$ or $c = b$, and 2 solutions if $c < a$ or $c > b$.

Part (iii)

Let $F(x) = |x - c| + |x - d|$. By part (i), F is a second bath, with floor at height $d - c$ over $[c, d]$ and walls of gradient ± 2. Since $d - c < b - a$, its floor is lower and narrower than that of f. Write

$$m = \tfrac{1}{2}(a + b), \qquad n = \tfrac{1}{2}(c + d)$$

for the midpoints of the two floors, and let $h(x) = f(x) - F(x)$.

Far to the left (for $x \leq \min\{a, c\}$) both graphs are on their left walls, so $h(x) = (a + b - 2x) - (c + d - 2x) = 2(m - n)$. Far to the right (for $x \geq \max\{b, d\}$), $h(x) = 2(n - m)$. Between these, h is continuous and piecewise linear.

> **?** **Ask yourself** Can the two graphs share a whole segment, giving infinitely many solutions?

Only where they have the same gradient. Both floors are flat, but they lie at different heights $b - a \neq d - c$, so they can't overlap. Otherwise the gradients agree only when both graphs are on their left walls ($x < \min\{a, c\}$) or both on their right walls ($x > \max\{b, d\}$), and there $h = \pm 2(m - n)$. So the graphs share a segment exactly when $m = n$.

Case $a + b = c + d$: infinitely many solutions. Now $h = 0$ for $x \leq \min\{a, c\}$ and $x \geq \max\{b, d\}$. Since $d - c < b - a$ and the midpoints agree, $a < c < d < b$: the small

bath sits centrally inside the big one. For $c \le x \le b$ the graph of F is on its floor or its right wall, so $F(x) \le 2b - c - d = b - a = f(x)$, with equality only at $x = b$. The same holds on $[a, d]$ by symmetry. So the solution set is exactly $x \le a$ together with $x \ge b$.

Case $a + b < c + d$: exactly one solution. Here $h = 2(m - n) < 0$ far to the left and $h = 2(n - m) > 0$ far to the right. Where can h decrease? We need the gradient of f to be less than that of F. There are two ways.

- f on its left wall while F is not, which needs $c < x < a$. Then $c < a$, and since $d - c < b - a$ also $d < b$, giving $c + d < a + b$, which is false here.
- f on its floor while F is on its right wall, which needs $d < x < b$.

So h is non-decreasing on $(-\infty, d]$. On $[d, \infty)$, h is non-increasing on $[d, b]$ (if $d < b$) and then constant at $2(n - m)$ for $x \ge \max\{b, d\}$. So $h(x) \ge 2(n - m) > 0$ for all $x \ge d$. On $(-\infty, d]$, h climbs from the value $2(m - n) < 0$ up to $h(d) > 0$ without decreasing, and it is not constant on any interval where it is zero (by the box above). So h has exactly one zero there. That gives exactly one solution in total.

Case $a + b > c + d$: exactly one solution, by the reflection $x \mapsto -x$, which replaces a, b, c, d by $-b, -a, -d, -c$ and reverses the inequality between the sums.

In summary, with $a < b$, $c < d$ and $d - c < b - a$: if $a + b = c + d$ (the two intervals share a midpoint) there are infinitely many solutions, namely every $x \le a$ and every $x \ge b$. Otherwise there is exactly one solution.

Checks

- Part (ii) with $a = 1$, $b = 3$, $c = 0$: the solutions $x = a + b - c = 4$ and $x = b - a + c = 2$ give $|3| + |1| = 4$ and $1 + 1 = 2$, as required. With $c = -5$ the second solution is $\frac{1}{3}(1 + 3 - 5) = -\frac{1}{3}$, and indeed $\frac{4}{3} + \frac{10}{3} = \frac{14}{3} = |-\frac{1}{3} + 5|$.
- Part (iii) with $a = 0$, $b = 10$, $c = 6$, $d = 7$: h is -3 on the far left, rises to 9 on $[6, 7]$, then falls back to 3 and stays there. Its only zero is $x = \frac{3}{2}$, where both sides equal 10.
- A computer check over tens of thousands of random integer choices of a, b, c, d found exactly the solution counts above in every case.

Common slips

- In (ii), forgetting the boundary cases $c = a$ and $c = b$, where the V just touches a corner of the square.
- In (iii), assuming h is monotonic. It isn't: when the small bath sits off-centre inside the big one, h decreases on (d, b). What saves the count is that h is already positive there.
- In (iii), missing the infinite case when the midpoints coincide.

Beyond STEP

The function $|x-a|+|x-b|$ is the total distance from x to two points. Minimising sums of distances like this is the idea behind the median in statistics and the ℓ^1 ("taxicab") norm. Functions like f are convex, and the arguments here about gradients increasing from piece to piece are the start of convex analysis, which underlies linear programming and optimisation. Bartle and Sherbert, *Introduction to Real Analysis*, makes the continuity and monotonicity arguments used here rigorous.

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